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(54) **INDEPENDENT WING POSITION CONTROL FOR AN AGRICULTURAL IMPLEMENT**

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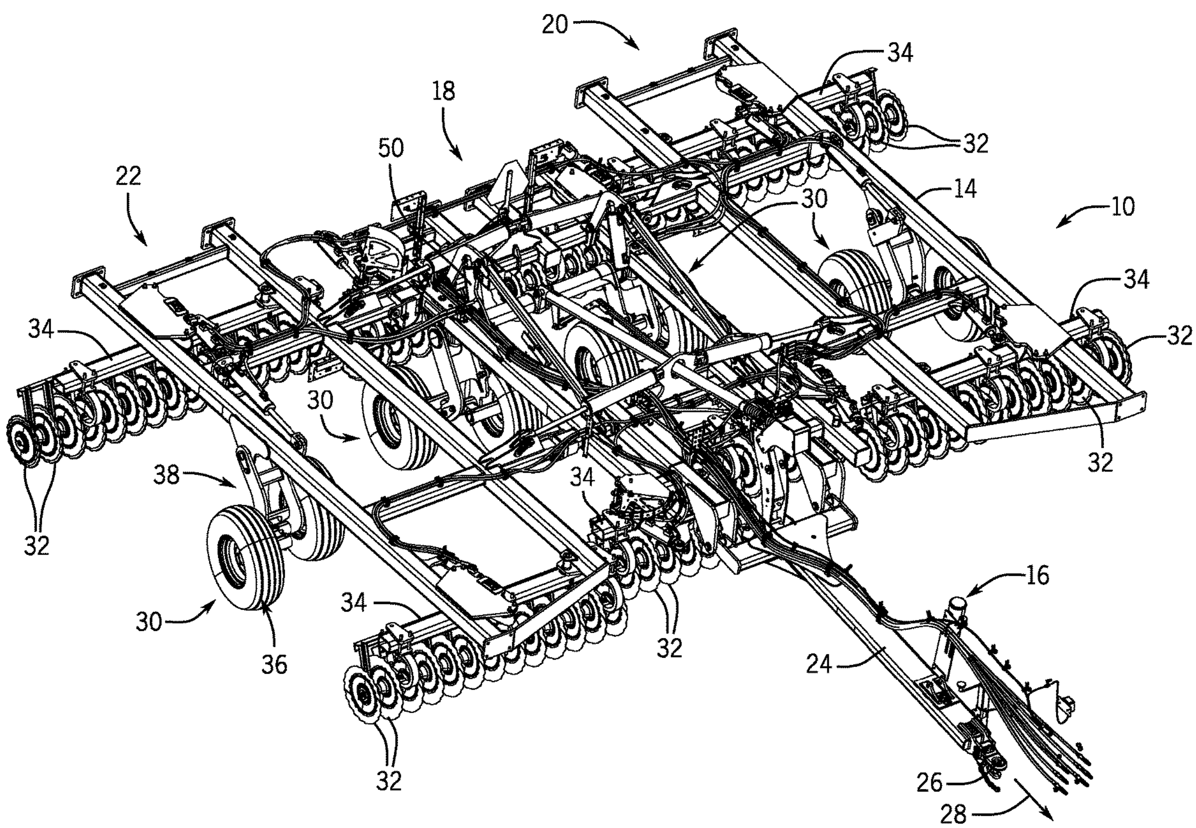
(57) **ABSTRACT**

Present embodiments disclose a height adjustment system for an agricultural implement that includes a main lift cylinder configured to control a vertical position of a center section wheel assembly, such that the center section wheel assembly is movably coupled to a center section of a frame of the agricultural implement, and a wing lift cylinder configured to control a vertical position of a wing section wheel assembly, such that the wing section wheel assembly is movably coupled to a wing section of the frame of the agricultural implement. Additionally, the height adjustment system for the agricultural implement includes a hydraulic circuit that includes a main lift metering valve configured to control hydraulic fluid flow to and from the main lift cylinder and the wing lift cylinder to control the vertical position of the center section wheel assembly and the vertical position of the wing section wheel assembly in unison.



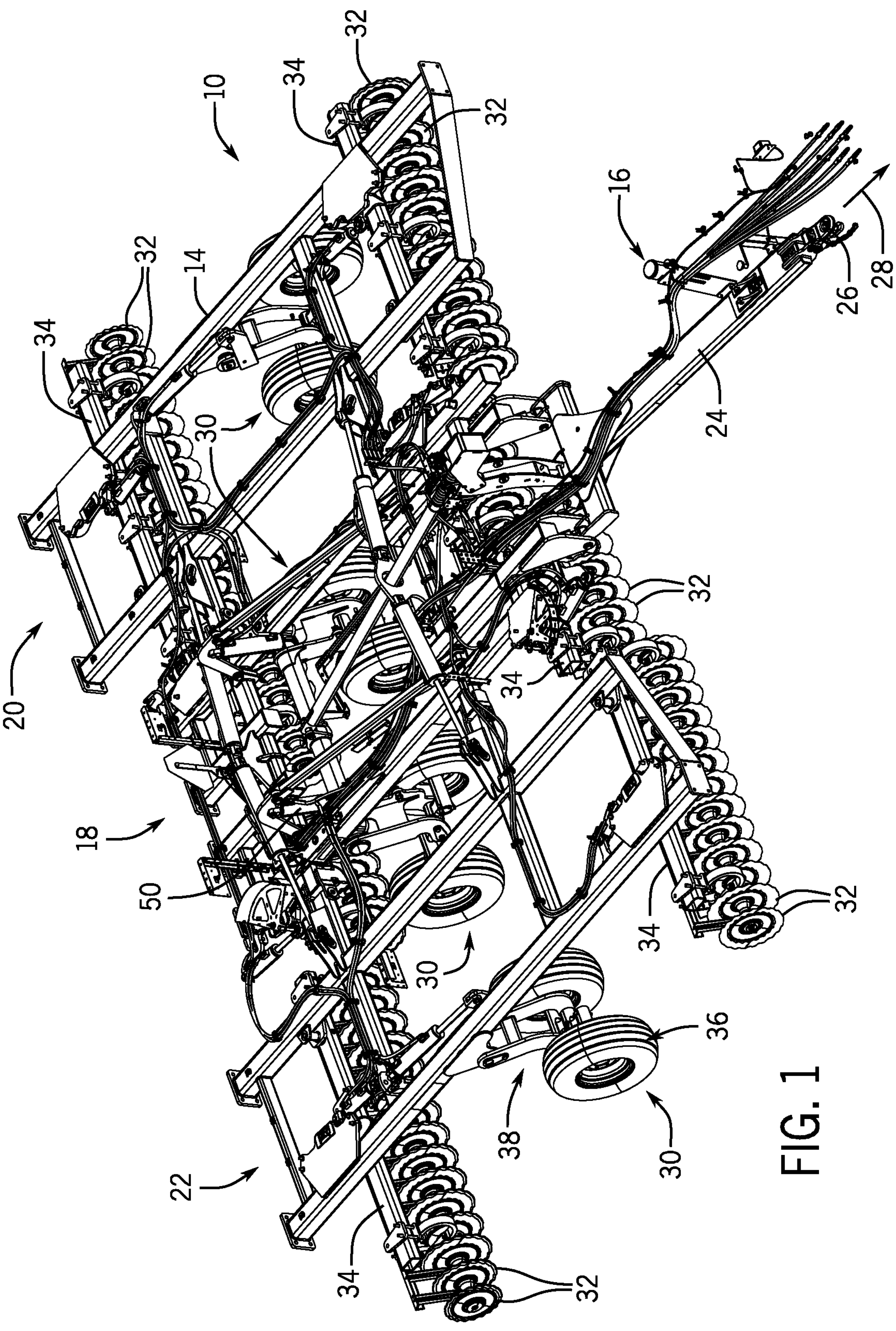


FIG. 1

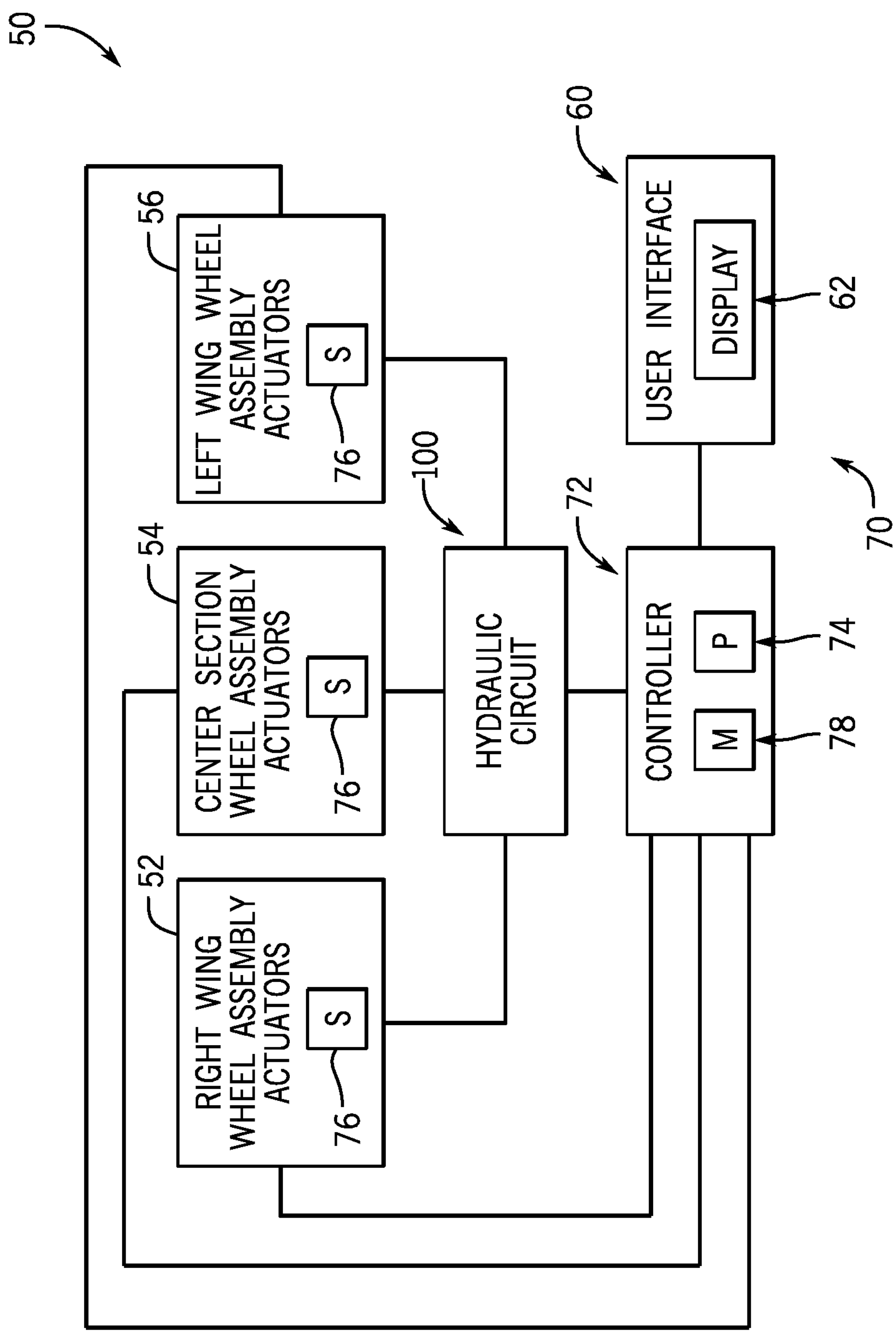
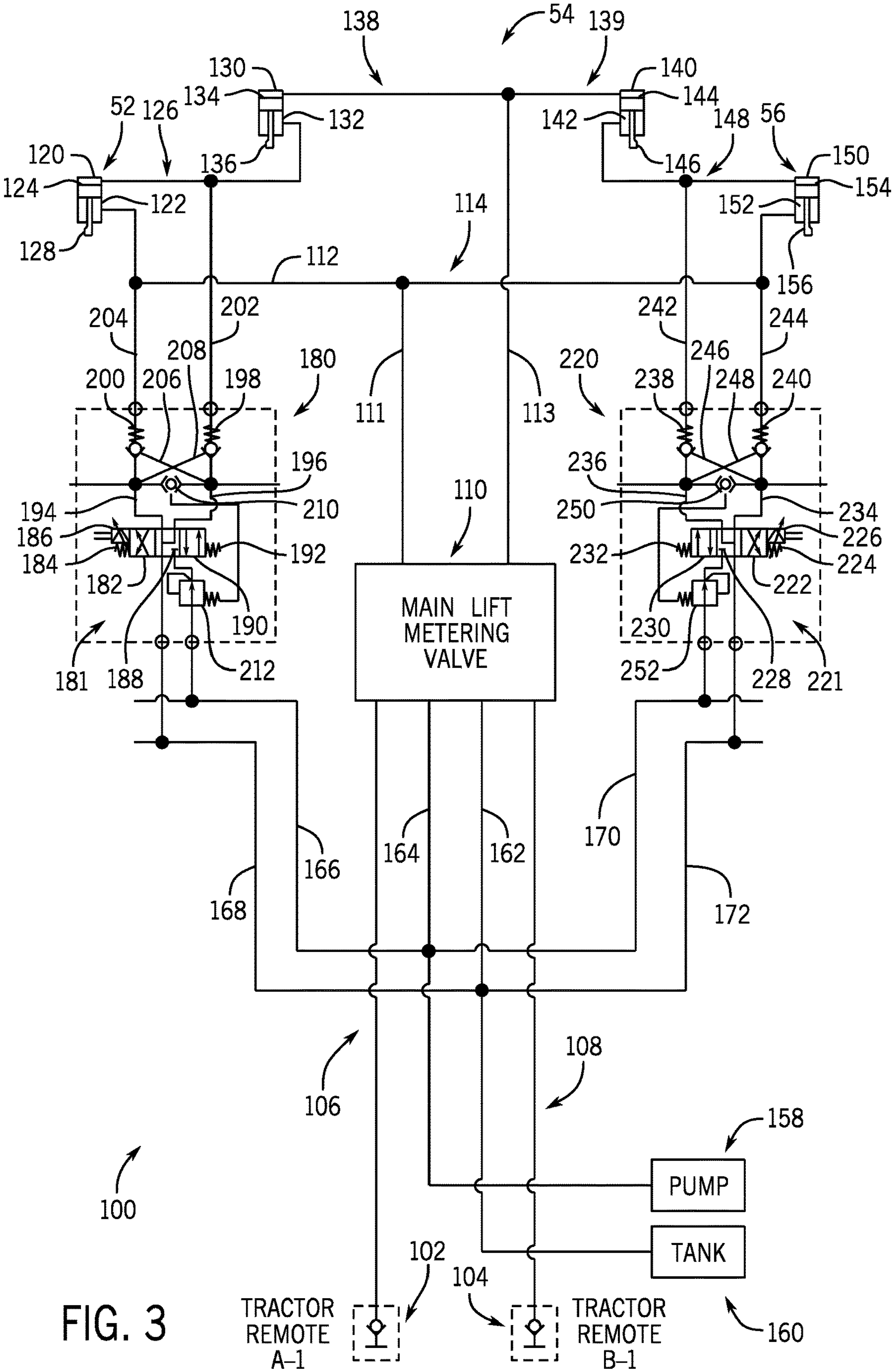

50
56
LEFT WING WHEEL ASSEMBLY ACTUATORS
S
76
54
CENTER SECTION WHEEL ASSEMBLY ACTUATORS
S
76
52
RIGHT WING WHEEL ASSEMBLY ACTUATORS
S
76
100
HYDRAULIC CIRCUIT
72
CONTROLLER
P
74
M
78
60
USER INTERFACE
DISPLAY
62
70

FIG. 2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220
238
248
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111
113
196
236
194
210
250
234
186
110
226
184
192
232
224
182
222
MAIN LIFT METERING VALVE
190
230
212
252
181
188
228
221
166
164
162
170
168
172
106
108
158
100
PUMP
TANK
102
104
TRACTOR REMOTE A–1
TRACTOR REMOTE B–1
160

FIG. 3

INDEPENDENT WING POSITION CONTROL FOR AN AGRICULTURAL IMPLEMENT

BACKGROUND

[0001] The present disclosure relates generally to independent wing position control for an agricultural implement.

[0002] This section is intended to introduce the reader to various aspects of art that may be related to various aspects of the present disclosure, which are described and/or claimed below. This discussion is believed to be helpful in providing the reader with background information to facilitate a better understanding of the various aspects of the present disclosure. Accordingly, it should be understood that these statements are to be read in this light, and not as admissions of prior art.

[0003] Certain agricultural implements include ground-engaging tools configured to interact with soil. For example, a tillage implement is an agricultural implement that may include multiple disc blades configured to break up the soil for subsequent planting and/or seeding operations. In operation, the tillage implement is coupled to a towing vehicle, such as a tractor, and as the tillage implement is towed behind the towing vehicle, the tillage implement prepares the soil by way of mechanical agitation. The disc blades may be arranged in groups, such as gangs, and coupled to a frame of the tillage implement in a configuration suitable for mechanically agitating the soil throughout a suitable penetration depth.

[0004] However, due to certain factors (e.g., uneven terrain, soil conditions, variable crop residue layers, etc.), the ground-engaging tools (e.g., disc blades) of the tillage implement may not suitably penetrate the soil to a desired penetration depth. For example, while towing the implement, an operator may notice that a portion of the tillage implement is tilling the soil at a different depth from an adjacent portion of the tillage implement. Uneven and variable soil tillage may lead to inconsistent soil conditions, which may result in inconsistent crop yield.

BRIEF DESCRIPTION

[0005] This summary is provided to introduce a selection of concepts that are further described below in the detailed description. This summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used as an aid in limiting the scope of the claimed subject matter.

[0006] In certain embodiments, a height adjustment system for an agricultural implement includes a main lift cylinder configured to control a vertical position of a center section wheel assembly, such that the center section wheel assembly is movably coupled to a center section of a frame of the agricultural implement, and a wing lift cylinder configured to control a vertical position of a wing section wheel assembly, such that the wing section wheel assembly is movably coupled to a wing section of the frame of the agricultural implement. Additionally, the height adjustment system for the agricultural implement includes a hydraulic circuit that includes a main lift metering valve configured to control hydraulic fluid flow to and from the main lift cylinder and the wing lift cylinder to control the vertical position of the center section wheel assembly and the vertical position of the wing section wheel assembly in unison, and a wing control valve circuit that includes a first control valve, such that the first control valve is configured to control fluid flow to and from the wing lift cylinder to control the vertical position of the wing section wheel assembly independently of the center section wheel assembly.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] These and other features, aspects, and advantages of the present disclosure will become better understood when the following detailed description is read with reference to the accompanying drawings in which like characters represent like parts throughout the drawings, wherein:

[0008] FIG. **1** is a perspective view of an embodiment of an agricultural implement having a height adjustment system, in accordance with aspects of the present disclosure;

[0009] FIG. **2** is a schematic view of an embodiment of a height adjustment system that may be employed within the agricultural implement of FIG. **1**, in accordance with aspects of the present disclosure; and

[0010] FIG. **3** is a schematic view of an embodiment of a hydraulic circuit that may be employed within the height adjustment system of FIG. **2**, in accordance with aspects of the present disclosure.

DETAILED DESCRIPTION

[0011] One or more specific embodiments of the present disclosure will be described below. In an effort to provide a concise description of these embodiments, all features of an actual implementation may not be described in the specification. It should be appreciated that in the development of any such actual implementation, as in any engineering or design project, numerous implementation-specific decisions must be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which may vary from one implementation to another. Moreover, it should be appreciated that such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure.

[0012] When introducing elements of various embodiments of the present disclosure, the articles "a," "an," "the," and "said" are intended to mean that there are one or more of the elements. The terms "comprising," "including," and "having" are intended to be inclusive and mean that there may be additional elements other than the listed elements. Any examples of operating parameters and/or environmental conditions are not exclusive of other parameters/conditions of the disclosed embodiments.

[0013] Farmers utilize a wide variety of agricultural equipment in order to prepare fields for subsequent agricultural operations. For example, a top layer of soil may be prepared for subsequent seeding or planting operations. To prepare the top layer of the soil, a tillage implement with a center section and multiple wing sections may be towed through a field by a work vehicle, such as a tractor. Ground-engaging tools (e.g., disc blades) are coupled to the center section and each wing section. Maintaining a suitably consistent penetration depth of the ground-engaging tools establishes consistent mechanical soil agitation throughout the field, thereby providing consistent crop yield after the subsequent seeding or planting operations. However, as the tillage implement is towed through the field, the ground-engaging tools of one of the sections (e.g., center section, left wing section, right wing section, etc.) may interact with the top layer of soil at a different penetration depth than the ground-engaging tools of neighboring sections.

[0014] Present embodiments are directed to facilitating and improving a tillage implement height adjustment system. For example, present embodiments include individual hydraulic control valve circuits corresponding to an individual section of the agricultural implement. These individual hydraulic control valve circuits may include control valves that are configured to enable a control system or an operator to independently adjust a vertical position of individual sections of the agricultural implement, in relation to a frame or the top layer of the soil. The control valves may be a portion of a hydraulic circuit, and may be fluidly coupled to a hydraulic cylinder, which in turn may be coupled to a wheel assembly. By introducing hydraulic fluid through the hydraulic circuit, the operator or control system may adjust the vertical position of the agricultural implement as an entire unit in unison, while also independently adjusting the vertical position of individual sections of the implement.

[0015] Present embodiments provide a user interface and display that enables the operator to input adjustments to the vertical positions of the individual sections of the implement. For example, the control system may be configured to communicatively couple with sensors on the individual sections of the implement, and enable the operator to adjust the vertical position of the individual sections of the implement based on the sensor feedback. By utilizing the control system and its components, the user may adjust the vertical position of individual sections of the implement during operation, thereby enabling consistent penetration depth by mechanical agitation tools coupled to each section of the implement. By utilizing the present embodiments of the implement height adjustment system, agricultural operations are made more efficient.

[0016] FIG. **1** is a perspective view of an embodiment of an agricultural implement (e.g., tillage implement) **10** having a height adjustment system **50**. In the illustrated embodiment, the agricultural implement is a tillage implement **10** (e.g., vertical tillage implement) having multiple ground-engaging tools configured to till soil. As discussed in further detail below, the height adjustment system **50** includes various hydraulic components and a control system that enables a user to adjust a vertical position of the tillage implement **10** during operation. As illustrated, the tillage implement **10** includes a frame **14** and a hitch assembly **16** coupled to the frame **14**. In the illustrated embodiment, the frame **14** includes a center section **18**, a left wing section **20**, and a right wing section **22**. Each wing section is configured to rotate upwardly from the illustrated working position to a transport position to facilitate transport of the tillage implement **10**. For example, one or more actuators (e.g. hydraulic cylinder(s), etc.) may be configured to drive each wing section to rotate between the illustrated working position and the transport position. While the frame **14** includes the center section **18**, the left wing section **20**, and the right wing section **22** in the illustrated embodiment, in other embodiments, the frame may include more or fewer sections. Furthermore, the frame **14** may be formed from multiple frame elements (e.g., rails, tubes, braces, etc.) coupled to one another (e.g., via welded connection(s), via fastener(s), etc.).

[0017] In the illustrated embodiment, the hitch assembly **16** includes a hitch frame **24** and a hitch **26**. The hitch frame **24** is pivotally coupled to the implement frame **14** via pivot joint(s), and the hitch **26** is configured to couple to a corresponding hitch of a work vehicle (e.g. tractor), which is configured to tow the tillage implement **10** through a field along a direction of travel **28**. While the hitch frame **24** is pivotally coupled to the implement frame **14** in the illustrated embodiment, in other embodiments, the hitch frame may be movably coupled to the implement frame by a linkage assembly (e.g. four bar linkage assembly, etc.) or another suitable assembly/mechanism that enables the hitch to move along a vertical axis relative to the implement frame, or the hitch frame may be rigidly coupled to the implement frame.

[0018] As illustrated, the tillage implement **10** includes wheel assemblies **30** movably coupled to the implement frame **14**. In the illustrated embodiment, each wheel assembly **30** includes a wheel frame **38** and a wheel **36** rotatably coupled to the wheel frame. The wheels **36** of the wheel assemblies **30** are configured to engage the surface of the soil, and the wheel assemblies **30** are configured to support at least a portion of the weight of the tillage implement **10**. In the illustrated embodiment, each wheel frame **38** is pivotally coupled to the implement frame **14**, thereby facilitating adjustment of the vertical position of the respective wheel(s) **36** relative to the implement frame **14**. However, in other embodiments, at least one wheel frame **38** may be movably coupled to the implement frame **14** by another suitable connection (e.g. sliding connection, linkage assembly, etc.) that facilitates adjustment of the vertical position of the respective wheel(s) **36** relative to the implement frame **14**. In a non-limiting embodiment, the height adjustment system **50** includes one or more actuators (e.g., hydraulic cylinder(s), etc.) that may be configured to adjust the vertical position of at least one wheel assembly **30** with respect to the frame **14**. In some embodiments, the one or more actuators are configured to adjust the vertical position of all wheel assemblies **30** coupled to the implement frame **14** concurrently and by a substantially similar amount. In addition, in certain embodiments, at least one of the actuators of the height adjustment system **50** may be configured to adjust the vertical position of at least one respective wheel assembly **30** individually with respect to the frame **14** and other wheel assemblies.

[0019] In the illustrated embodiment, the tillage implement **10** includes disc blades **32** configured to engage a top layer of the soil. As the tillage implement **10** is towed through the field, the disc blades **32** are driven to rotate, thereby breaking up the top layer of the soil. In the illustrated embodiment, the disc blades **32** are arranged in two rows. However, in other embodiments, the disc blades may be arranged in more or fewer rows (e.g. 1, 3, 4, 5, 6, or more). Furthermore, in the illustrated embodiment, each row of disc blades **32** includes four gangs of disc blades **32**. Two gangs of disc blades of the front row are coupled to the center section **18**, two gangs of disc blades of the rear row are coupled to the center section **18**, one gang of disc blades of the front row is coupled to the left wing section **20**, one gang of disc blades of the rear row is coupled to the left wing section **20**, one gang of disc blades of the front row is coupled to the right wing section **22**, and one gang of disc blades of the rear row is coupled to the right wing section **22**. While the tillage implement **10** includes eight gangs of disc blades **32** in the illustrated embodiment, in other embodiments, the tillage implement may include more or fewer gangs of disc blades (e.g. 2, 4, 6, 10, or more). Furthermore, the gangs of disc blades may be arranged in any suitable configuration on the implement frame.

[0020] The disc blades **32** of each gang are non-rotatably coupled to one another by a respective shaft, such that the disc blades **32** of each gang rotate together. Each shaft is rotatably coupled to a respective disc blade support **34**, which is configured to support the gang, including the shaft and the disc blades **32**. Furthermore, each disc blade support **34** is pivotally coupled to the frame **14** at a respective pivot point, thereby enabling the disc blade support **34** to rotate relative to the frame **14**. Rotating the disc blade support **34** relative to the frame **14** controls the angle between the respective disc blades **32** and the direction of travel **28**, thereby controlling the interaction of the disc blades **32** with the top layer of the soil. While each disc blade support **34** is pivotally coupled to the frame **14** in the illustrated embodiment, in other embodiments, at least one disc blade support (e.g., each disc blade support) may be fixed to the frame, such that the angle of the disc blades relative to the direction of travel is fixed.

[0021] In a non-limiting embodiment, at least a portion of the interaction of the disc blades **32** with the top layer of the soil is controlled by the vertical position of the aforementioned wheel assemblies **30**. As used herein with regard to the wheel assemblies, "vertical position" refers to a vertical position/height of the wheel(s) of the wheel assembly relative to the respective section of the implement frame. Controlling the vertical position of the wheel assemblies **30** controls the height of the frame **14** above the ground, thereby controlling the penetration depth of the disc blades **32**. In addition, because each wing section of the frame **14** is pivotally coupled to the center section **18** of the frame **14**, controlling the vertical position of wheel assembly/assemblies **30** coupled to a wing section controls the penetration depth of the disc blades **32** coupled to the wing section independently of the disc blades **32** coupled to the center section **18**. For example, if the wheel assembly/assemblies **30** coupled to the left wing section **20** are actuated to a lower vertical position, then the gang(s) of disc blades **32** coupled to the left wing section **20** may have a shallower interaction with the top layer of soil. Additionally, if the wheel assembly/assemblies **30** coupled to the left wing section **20** are actuated to a higher vertical position, then the gang(s) of disc blades **32** coupled to the left wing section **20** may have a deeper interaction with the top layer of soil.

[0022] Each disc blade support **34** may include any suitable structure(s) configured to support the respective gang (e.g. including a square tube, a round tube, a bar, a truss, other suitable structure(s), or a combination thereof). While the disc blades **32** supported by each disc blade support **34** are arranged in a respective gang (e.g. non-rotatably coupled to one another by a respective shaft) in the illustrated embodiment, in other embodiments, at least a portion of the disc blades supported by at least one disc blade support (e.g. all of the disc blades supported by the disc blade support) may be arranged in another suitable configuration (e.g., individually mounted and independently rotatable, mounted in groups and individually rotatable, etc.). For example, in certain embodiments, a first portion of the disc blades supported by a disc blade support may be arranged in a gang, and a second portion of the disc blades supported by the disc blade support may be individually mounted and independently rotatable.

[0023] While the tillage implement **10** includes the disc blades **32** in the illustrated embodiment, in other embodiments, the tillage implement may include other/additional ground engaging tool(s) (e.g., coupled to the disc blade support(s), coupled to the frame of the tillage implement, etc.). For example, in certain embodiments, the tillage implement may include tillage point assemblies (e.g., positioned behind the disc blades relative to the direction of travel) configured to engage the soil at a greater depth than the disc blades, thereby breaking up a lower layer of soil. Each tillage point assembly may include a tillage point and a shank. The shank may position the tillage point at a target depth beneath the soil surface, and the tillage point may break up the soil. The shape of each tillage point, the arrangement of the tillage point assemblies, and the number of tillage point assemblies may be selected to control tillage within the field. Furthermore, in certain embodiments, the tillage implement may include finishing discs (e.g. positioned behind the disc blades relative to the direction of travel). In such embodiments, as the tillage implement is towed through the field, the finishing discs may be driven to rotate, thereby sizing soil clods, leveling the soil surface, smoothing the soil surface, cutting residue on the soil surface, or a combination thereof. In addition, in certain embodiments, the tillage implement may include one or more other/additional suitable ground engaging tools, such as coulter(s), opener(s), tine(s), finishing reel(s), other suitable ground engaging tool(s), or a combination thereof. Furthermore, while the tillage implement is a vertical tillage implement in the illustrated embodiment, in other embodiments, the tillage implement may be a primary tillage implement or another suitable type of tillage implement.

[0024] FIG. **2** is a schematic view of a height adjustment system **50** that may be employed within the agricultural implement **10** of FIG. **1**. The height adjustment system **50** includes right wing wheel assembly actuator(s) **52**, center section wheel assembly actuator(s) **54**, left wing wheel assembly actuator(s) **56**, a control system **70**, a hydraulic circuit **100**, and a user interface **60**. The right wing wheel assembly actuator(s) **52** are configured to control the vertical position of the right wing wheel assembly/assemblies, the center section wheel assembly actuator(s) **54** are configured to control the vertical position of the center section wheel assembly/assemblies, and the left wing wheel assembly actuator(s) **56** are configured to control the vertical position of the left wing wheel assembly/assemblies. Additionally, the right wing wheel assembly actuator(s) **52**, the center section wheel assembly actuator(s) **54**, and the left wing wheel assembly actuator(s) **56** are fluidly coupled to components of the hydraulic circuit **100**, as discussed in further detail below.

[0025] As discussed previously, an individual right wing wheel assembly includes a wheel frame and a suitable number of wheel(s) rotatably coupled to the wheel frame. In certain embodiments, one (1) right wing wheel assembly is coupled to the right wing section, with the right wing wheel assembly including one (1) wheel frame and two (2) wheels. However, in other embodiments, more than one right wing wheel assembly (2, 3, 4, etc.) may be coupled to the right wing section, and any suitable number of right wing wheel assemblies is considered within the scope of the various embodiments of the present disclosure. Additionally or alternatively, the right wing wheel assembly may include a single wheel or more than two (2) wheels, and these configurations are considered within the scope of the present disclosure. The right wing wheel assembly is configured to provide support for the right wing section and to enable the right wing section to travel through the field as the tillage implement is being towed in the direction of travel.

[0026] In a non-limiting embodiment, the wheel frame of the right wing wheel assembly is pivotally coupled to the right wing section of the frame, and the wheel frame is mechanically coupled to right wing wheel assembly actuator (s) **52**. The right wing wheel assembly is configured to pivot between a first position and a second position based on extension of the right wing wheel assembly actuator(s) **52**. In a non-limiting embodiment, the hydraulic circuit **100** may introduce specific volumes of hydraulic fluid to the rod end(s) or the cap end(s) of the right wing wheel assembly actuator(s) **52**, thereby driving the right wing wheel assembly to rotate. As a result, the right wing wheel assembly may position the respective wheel(s) in a fully lowered position (e.g., corresponding to the first position of the right wing wheel assembly), a fully raised position (e.g., corresponding to the second position of the right wing wheel assembly), as well as intermediate positions between the fully lowered position and the fully raised position.

[0027] As it relates to the operation of the tillage implement, when the right wing wheel assembly is in the first position, the vertical position of the right wing section is at a maximum. When the vertical position of the right wing section is at a maximum, the corresponding disc blades coupled to the right wing section are disengaged from the top layer of the soil. On the other hand, when the right wing wheel assembly is in the second position, the vertical position of the right wing section is at a minimum. When the vertical position of the right wing section is at a minimum, the corresponding disc blades coupled to the right wing section are at a maximum penetration depth within the top layer of the soil. Additionally, when the right wing wheel assembly is in one of the intermediate positions, the vertical position of the right wing section is in an intermediate position. When the vertical position of the right wing section is in one of the intermediate positions, the corresponding disc blades coupled to the right wing section may be engaged with the top layer of soil at a penetration depth or disengaged with the top layer of soil.

[0028] Turning to the middle of the agricultural implement **10**, an individual center section wheel assembly includes a wheel frame and a suitable number of wheel(s) rotatably coupled to the wheel frame. In certain embodiments, two (2) center section wheel assemblies are coupled to the center section, with the center section wheel assemblies each including one (1) wheel frame and two (2) wheels. However, in other embodiments, fewer or greater than two center section wheel assemblies (1, 3, 4, etc.), and any suitable number of center section wheel assemblies is considered within the scope of the various embodiments of the present disclosure. Additionally or alternatively, the center section wheel assemblies may include a single wheel or more than two (2) wheels, and these configurations are considered within the scope of the present disclosure. The center section wheel assemblies are configured to provide support for the center section and to enable the center section to travel through the field as the tillage implement is being towed in the direction of travel.

[0029] In a non-limiting embodiment, the wheel frame of the center section wheel assemblies are pivotally coupled to the center section of the frame, and the wheel frame is mechanically coupled to center section wheel assembly actuator(s) **54**. The center section wheel assemblies are configured to pivot between a third position and a fourth position based on extension of each center section wheel assembly actuator(s) **54**. In a non-limiting embodiment, the hydraulic circuit **100** may introduce specific volumes of hydraulic fluid to the rod end(s) or the cap end(s) of the center section wheel assembly actuator(s) **54**, thereby driving the center section wheel assemblies to rotate. As a result, the center section wheel assemblies may position the respective wheel(s) in a fully lowered position (e.g., corresponding to the third position of the center section wheel assemblies), a fully raised position (e.g., corresponding to the fourth position of the center section wheel assemblies), as well as intermediate positions between the fully lowered position and the fully raised position.

[0030] As it relates to the operation of the tillage implement, when the center section wheel assemblies are in the third position, the vertical position of the center section is at a maximum. When the vertical position of the center section is at a maximum, the corresponding disc blades coupled to the center section are disengaged from the top layer of the soil. On the other hand, when the center section wheel assemblies are in the fourth position, the vertical position of the center section is at a minimum. When the vertical position of the center section is at a minimum, the corresponding disc blades coupled to the center section are at a maximum penetration depth within the top layer of the soil. Additionally, when the center section wheel assemblies are in one of the intermediate positions, the vertical position of the center section is in an intermediate position. When the vertical position of the center section is in one of the intermediate positions, the corresponding disc blades coupled to the center section may be engaged with the top layer of soil at a penetration depth or disengaged with the top layer of soil.

[0031] As discussed previously, an individual left wing wheel assembly includes a wheel frame and a suitable number of wheel(s) rotatably coupled to the wheel frame. In certain embodiments, one (1) left wing wheel assembly is coupled to the left wing section, with the left wing wheel assembly including one (1) wheel frame **38** and two (2) wheels **36**. However, in other embodiments, more than one left wing wheel assembly (2, 3, 4, etc.), and any suitable number of left wing wheel assemblies is considered within the scope of the various embodiments of the present disclosure. Additionally or alternatively, the left wing wheel assembly may include a single wheel **36** or more than two (**2**) wheels, and these configurations are considered within the scope of the present disclosure. The left wing wheel assembly is configured to provide support for the left wing section and to enable the left wing section to travel through the field as the tillage implement is being towed in the direction of travel.

[0032] In a non-limiting embodiment, the wheel frame of the left wing wheel assembly is pivotally coupled to the left wing section of the frame, and the wheel frame is mechanically coupled to left wing wheel assembly actuator(s) **56**.

The left wing wheel assembly is configured to pivot between a fifth position and a sixth position based on extension of the left wing wheel assembly actuator(s) **56**. In a non-limiting embodiment, the hydraulic circuit **100** may introduce specific volumes of hydraulic fluid to the rod end(s) or the cap end(s) of the left wing wheel assembly actuator(s) **56**, thereby driving the left wing wheel assembly to rotate. As a result, the left wing wheel assembly may position the respective wheel(s) in a fully lowered position (e.g., corresponding to the fifth position of the left wing wheel assembly), a fully raised position (e.g., corresponding to the sixth position of the left wing wheel assembly), as well as intermediate positions between the fully lowered position and the fully raised position.

[0033] As it relates to the operation of the tillage implement, when the left wing wheel assembly is in the fifth position, the vertical position of the left wing section is at a maximum. When the vertical position of the left wing section is at a maximum, the corresponding disc blades coupled to the left wing section are disengaged from the top layer of the soil. On the other hand, when the left wing wheel assembly is in the sixth position, the vertical position of the left wing section is at a minimum. When the vertical position of the left wing section is at a minimum, the corresponding disc blades coupled to the left wing section are at a maximum penetration depth within the top layer of the soil. Additionally, when the left wing wheel assembly is in one of the intermediate positions, the vertical position of the left wing section is in an intermediate position. When the vertical position of the left wing section is in one of the intermediate positions, the corresponding disc blades coupled to the left wing section may be engaged with the top layer of soil at a penetration depth or disengaged with the top layer of soil.

[0034] The height adjustment system **50** also includes a control system **70** that is configured to receive input from the user interface **60** and control the hydraulic circuit **100** (e.g., based on the received input). The control system **70** includes a controller **72** that is communicatively coupled to the hydraulic circuit **100**. In a non-limiting embodiment, the controller **72** is configured to receive feedback from sensors disposed on the wheel assemblies. In such an embodiment, the sensors are communicatively coupled to the controller **72**. Further, the controller **72** may be configured to control the hydraulic circuit **100** based on the information received from the sensor feedback (e.g., from the sensors disposed on the wheel assemblies). Additionally or alternatively, the controller **72** may be configured to control the hydraulic circuit **100** based on the input received from the user interface **60**.

[0035] In the illustrated embodiment, the controller **72** includes a memory **78** and a processor **74**. The controller **72** may also include one or more storage devices and/or other suitable components. The processor **74** may be used to execute software, such as software for controlling the vertical position of the wheel assemblies in unison. In a non-limiting embodiment, the processor **74** may be used to execute software that enables the controller to control the vertical position of the right wing wheel assembly independently from the center section wheel assembly and the left wing wheel assembly. Additionally or alternatively, the processor **74** may be used to execute software that enables the controller to control the vertical position of the left wing wheel assembly independently from the center section wheel assembly and the right wing wheel assembly. Moreover, the processor **74** may include multiple microprocessors, one or more "general-purpose" microprocessors, one or more special-purpose microprocessors, and/or one or more application specific integrated circuits (ASICs), or some combination thereof. For example, the processor **74** may include one or more reduced instruction set (RISC) processors.

[0036] The memory **78** may include a volatile memory, such as random-access memory (RAM), and/or a nonvolatile memory, such as read-only memory (ROM). The memory **78** may store a variety of information and may be used for various purposes. For example, the memory **78** may store processor-executable instructions (e.g., firmware or software) for the processor **74** to execute, such as instructions for controlling the vertical position of the wheel assemblies in unison. In a non-limiting embodiment, the memory **78** may store processor-executable instructions related to controlling the vertical position of the right wing wheel assembly independently from the center section wheel assembly and left wing wheel assembly. Additionally or alternatively, the memory may store processor-executable instructions related to controlling the vertical position of the left wing wheel assembly independently from the center section wheel assembly and the right wing wheel assembly. The storage device(s) (e.g., nonvolatile storage) may include ROM, flash memory, a hard drive, or any other suitable optical, magnetic, or solid-state storage medium, or a combination thereof. The storage device(s) may store data, instructions (e.g., software or firmware for controlling the vertical position of the wheel assemblies), and any other suitable data.

[0037] In some embodiments, the controller **72** may control the vertical position of the right wing wheel assembly, the center section wheel assemblies, and the left wing wheel assembly based on sensor feedback. For example, one or more sensors **76** may be positioned at one or more wheel assemblies. The sensor(s) **76** may include a pressure sensor, a position sensor, a height sensor, and the like. For example, a first pressure sensor may be fluidly coupled to the right wing wheel assembly actuator(s) **52**, a second pressure sensor may be fluidly coupled to the center section wheel assembly actuator(s) **54**, and a third pressure sensor may be fluidly coupled to the left wing wheel assembly actuator(s) **56**. Each pressure sensor may monitor and output a signal indicative of a fluid pressure within the respective actuator (s). The controller **72** may control the hydraulic circuit **100** based on feedback from the pressure sensor(s) to control the vertical position of the corresponding wheel assembly/assemblies.

[0038] Position sensor(s) may be coupled to the respective wheel frame(s) of one or more wheel assemblies. For example, a first position sensor may be coupled to the wheel frame of the right wing wheel assembly, a second position sensor may be coupled to the wheel frame of a center section wheel assembly, and a third position sensor may be coupled to the wheel frame of the left wing wheel assembly. Each position sensor may monitor and output a signal corresponding to the vertical position of the respective wheel assembly. The controller **72** may control the hydraulic circuit **100** based on feedback from the position sensor(s) to control the vertical position of the corresponding wheel assembly/assemblies.

[0039] Additionally or alternatively, height sensor(s) may be coupled to one or more frame sections. Each height sensor may monitor and output a signal indicative of the height of the respective frame section above the ground, which is indicative of the penetration depth of the ground-engaging tools (e.g., disc blades, tillage point assemblies, finishing discs, coulter(s), opener(s), tine(s), finishing reel (s), other suitable ground engaging tool(s), etc.) coupled to the frame section. The controller **72** may control the hydraulic circuit **100** to adjust the vertical position of the corresponding wheel assembly/assemblies (e.g., the right wing wheel assembly, the center section wheel assemblies, the left wing wheel assembly, etc.) based on feedback from the height sensor(s) to adjust the penetration depth of the respective ground-engaging tools.

[0040] In the illustrated embodiment, the user interface **60** is communicatively coupled to the controller **72**. The user interface **60** is configured to receive input from an operator and to provide information to the operator (e.g., hydraulic fluid pressure(s) and/or flow rate(s), wheel assembly vertical position(s), etc.). The user interface **60** may include any suitable input device(s) for receiving input, such as a keyboard, a mouse, button(s), switch(es), knob(s), other suitable input device(s), or a combination thereof. In addition, the user interface **60** may include any suitable output device(s) for presenting information to the operator, such as speaker (s), indicator light(s), other suitable output device(s), or a combination thereof. In the illustrated embodiment, the user interface **60** includes a display **62** configured to present visual information to the operator (e.g., real time/near real time information, such as hydraulic fluid pressure(s) and/or flow rate(s), wheel assembly vertical position(s), etc.). In a non-limiting embodiment, the display **62** may include a touchscreen interface configured to receive input from the operator (e.g., a capacitive touch screen with haptic feedback).

[0041] In certain embodiments, the user interface **60** is configured to receive input from the operator indicative of a command to adjust the vertical position of the right wing wheel assembly, the center section wheel assemblies, and the right wing wheel assembly in unison. In certain embodiments, the user interface **60** is configured to enable the operator to input a command to adjust the vertical position of the right wing wheel assembly independently from the other wheel assemblies. Additionally or alternatively, in certain embodiments, the user interface **60** is configured to enable the operator to input a command to adjust the vertical position of the left wing wheel assembly independently from the other wheel assemblies. Furthermore, in certain embodiments, the user interface **60** may enable the operator to input a target vertical position corresponding to a desired setpoint for the vertical position of the wheel assemblies, an adjustment distance to the vertical position for each wing wheel assembly, other suitable parameter(s) (e.g., target height of each section above the ground, etc.), or a combination thereof. The controller **72** is configured to receive the inputs from the user interface **60** and to control the vertical position (s) of the wheel assembly/assemblies based on the inputs received from the operator via the user interface **60**.

[0042] In certain embodiments, the controller **72** is configured to output information to the user interface **60** corresponding to the sensor feedback received from the sensor (s) **76**. For example, in response to receiving signal(s) from one or more of the pressure sensors indicative of the fluid pressure being above or below a threshold fluid pressure, the controller **72** may instruct the user interface **60** to present a corresponding indication to the operator. Additionally or alternatively, in response to receiving signal(s) from one or more of the position sensors indicative of the position of the respective wheel frame(s) being above or below a threshold minimum or maximum position, the controller **72** may instruct the user interface **60** to present a corresponding indication to the operator. Furthermore, in response to receiving signal(s) from one or more of the height sensors indicative of the penetration depth of one or more of the ground-engaging tools being above or below a penetration depth threshold, the controller **72** may instruct the user interface **60** to present a corresponding indication to the operator. The user interface **60** is configured to provide the operator with the information from the controller **72** (e.g., via the display **62**). For example, in response to an indication on the user interface **60**, the operator may input, via the user interface **60**, an input to adjust the vertical position of one or more wheel assemblies.

[0043] FIG. **3** is a schematic diagram of an embodiment of a hydraulic circuit **100** that may be employed within the height adjustment system of FIG. **2**. In the illustrated embodiment, the hydraulic circuit **100** is configured to control the vertical positions of the wheel assemblies with respect to the frame of the tillage implement. The hydraulic circuit **100** includes a first hydraulic fluid input **102**, a second hydraulic fluid input **104**, a main lift metering valve **110**, a right wing control valve circuit **180**, and a left wing control valve circuit **220**. Furthermore, the right wing wheel assembly actuator(s) **52** include a single right wing lift cylinder **120**, the center section wheel assembly actuator(s) **54** include a right main lift cylinder **130** and a left main lift cylinder **140**, and the left wing wheel assembly actuator(s) **56** include a single left wing lift cylinder **150**. The first hydraulic fluid input **102** is configured to provide hydraulic fluid through a first supply conduit **106** to the main lift metering valve **110**. In addition, the second hydraulic fluid input **104** is configured to provide hydraulic fluid through a second supply conduit **108** to the main lift metering valve **110**. During operation, the main lift metering valve **110** is configured to receive hydraulic fluid from one of the first hydraulic fluid input **102** or the second hydraulic fluid input **104**, and hydraulic fluid may drain through the other hydraulic fluid input. For example, a tractor remote within the work vehicle, which is fluidly coupled to the hydraulic fluid inputs, may be controlled to selectively supply hydraulic fluid to the first hydraulic fluid input **102** or to the second hydraulic fluid input **104**, while enabling hydraulic fluid to drain through the other hydraulic fluid input.

[0044] The main lift metering valve **110** may receive hydraulic fluid from the first supply conduit **106** and output hydraulic fluid through the retraction conduit **111**. The retraction conduit **111** is fluidly connected to a right retraction branch **112** and fluidly connected to a left retraction branch **114**, such that hydraulic fluid output from the main lift metering valve **110** through the retraction conduit **111** is split between the right branch **112** and the left branch **114**. In addition, the right branch retraction conduit **112** is configured to provide hydraulic fluid to a rod end **122** of the right wing lift cylinder **120**, and the left branch retraction conduit **114** is configured to provide hydraulic fluid to a rod end **152** of the left wing lift cylinder **150**.

[0045] In addition, the main lift metering valve **110** may receive hydraulic fluid from the second supply conduit **108** and output hydraulic fluid through the extension conduit

113. The extension conduit **113** is fluidly connected to a right branch extension conduit **138** and fluidly connected to a left branch extension conduit **139**, such that hydraulic fluid output from the main lift metering valve **110** through the extension conduit **113** is split between the right branch **138** and the left branch **139**. In addition, the right branch extension conduit **138** is configured to provide hydraulic fluid to a cap end **134** of the right main lift cylinder **130**, and the left branch extension conduit **139** is configured to provide hydraulic fluid to a cap end **144** of the left main lift cylinder **140**.

[0046] In the illustrated embodiment, the right wing lift cylinder **120** includes a rod end **122**, a cap end **124**, and an actuating piston rod **128**. The rod end **122** of the right wing lift cylinder **120** is fluidly coupled to the right branch retraction conduit **112** and is configured to receive hydraulic fluid from the right branch retraction conduit **112**. When the right wing lift cylinder **120** receives hydraulic fluid into the rod end **122**, the hydraulic fluid drives the piston rod **128** to retract. In addition, the cap end **124** of the right wing lift cylinder **120** is fluidly coupled to a first cylinder conduit **126** that connects the cap end **124** of the right wing lift cylinder **120** to a rod end **132** of the right main lift cylinder **130**. In the illustrated embodiment, when the hydraulic fluid in the rod end **122** of the right wing lift cylinder **120** drives the piston rod **128** to retract, the piston rod **128** drives the hydraulic fluid present in the cap end **124** of the right wing lift cylinder **120** out of the cap end **124** and into the first cylinder conduit **126**. As discussed in more detail below, when hydraulic fluid is received into the cap end **124** via the first cylinder conduit **126**, the piston rod **128** is driven to extend, thereby driving hydraulic fluid present in the rod end **122** of the right wing lift cylinder **120** into the right branch retraction conduit **112**.

[0047] As discussed previously, the first cylinder conduit **126** fluidly connects the cap end **124** of the right wing lift cylinder **120** and the rod end **132** of the right main lift cylinder **130**. In a non-limiting embodiment, the right main lift cylinder **130** includes the rod end **132**, a cap end **134**, and an actuating piston rod **136**. The rod end **132** of the right main lift cylinder **130** is fluidly coupled to the first cylinder conduit **126** and is configured to receive hydraulic fluid from the first cylinder conduit **126**. When the right main lift cylinder **130** receives hydraulic fluid into the rod end **132**, the hydraulic fluid drives the piston rod **136** to retract. In addition, the cap end **134** of the right main lift cylinder **130** is fluidly coupled to the right branch extension conduit **138** that connects the cap end **134** of the right main lift cylinder **130** to the extension conduit **113**. In the illustrated embodiment, when the hydraulic fluid in the rod end **132** of the right main lift cylinder **130** drives the piston rod **136** to retract, the piston rod **136** drives the hydraulic fluid present in the cap end **134** of the right main lift cylinder **130** out of the cap end **134** and into the right branch extension conduit **138**, then to the extension conduit **113**, and then to the main lift metering valve **110**. As discussed in more detail below, when hydraulic fluid is received into the cap end **134** via the right branch extension conduit **138**, the piston rod **136** is driven to extend, thereby driving hydraulic fluid present in the rod end **132** of the right main lift cylinder **130** into the first cylinder conduit **126**.

[0048] In the illustrated embodiment, the left wing lift cylinder **150** includes a rod end **152**, a cap end **154**, and an actuating piston rod **156**. The rod end **152** of the left wing lift cylinder **150** is fluidly coupled to the left branch retraction conduit **114** and is configured to receive hydraulic fluid from the left branch retraction conduit **114**. When the left wing lift cylinder **150** receives hydraulic fluid into the rod end **152**, the hydraulic fluid drives the piston rod **156** to retract. In addition, the cap end **154** of the left wing lift cylinder **150** is fluidly coupled to a second cylinder conduit **148** that connects the cap end **154** of the left wing lift cylinder **150** to the rod end **142** of the left main lift cylinder **140**. In the illustrated embodiment, when the hydraulic fluid in the rod end **152** of the left wing lift cylinder **150** drives the piston rod **156** to retract, the piston rod **156** drives the hydraulic fluid present in the cap end **154** of the left wing lift cylinder **150** out of the cap end **154** and into the second cylinder conduit **148**. As discussed in more detail below, when hydraulic fluid is received into the cap end **154** via the second cylinder conduit **148**, the piston rod **156** is driven to extend, thereby driving hydraulic fluid present in the rod end **152** of the left wing lift cylinder **150** into the left branch retraction conduit **114**.

[0049] As discussed previously, the second cylinder conduit **148** fluidly connects the cap end **154** of the left wing lift cylinder **150** and the rod end **142** of the left main lift cylinder **140**. In a non-limiting embodiment, the left main lift cylinder **140** includes the rod end **142**, a cap end **144**, and an actuating piston rod **146**. The rod end **142** of the left main lift cylinder **140** is fluidly coupled to the second cylinder conduit **148** and is configured to receive hydraulic fluid from the second cylinder conduit **148**. When the left main lift cylinder **140** receives hydraulic fluid into the rod end **142**, the hydraulic fluid drives the piston rod **146** to retract. In addition, the cap end **144** of the left main lift cylinder **140** is fluidly coupled to the left branch extension conduit **139** that connects the cap end **144** of the left main lift cylinder **140** to the extension conduit **113**. In the illustrated embodiment, when the hydraulic fluid in the rod end **142** of the left main lift cylinder **140** drives the piston rod **146** to retract, the piston rod **146** drives the hydraulic fluid present in the cap end **144** of the left main lift cylinder **140** out of the cap end **144** and into the left branch extension conduit **139**, then to the extension conduit **113**, and then to the main lift metering valve **110**. As discussed in more detail below, when hydraulic fluid is received into the cap end **144** via the left branch extension conduit **139**, the piston rod **146** is driven to extend, thereby driving hydraulic fluid present in the rod end **142** of the left main lift cylinder **140** into the second cylinder conduit **148**.

[0050] In a non-limiting embodiment, the frame of the right wing wheel assembly is coupled to the right wing lift cylinder **120**, the frame of a first center section wheel assembly is coupled to the right main lift cylinder **130**, the frame of a second center section wheel assembly is coupled to the left main lift cylinder **140**, and the frame of the left wing wheel assembly is coupled to the left wing lift cylinder **150**. By providing hydraulic fluid to the first hydraulic fluid input **102**, hydraulic fluid flows through the first supply conduit **106** to the main lift metering valve **110**. The main lift metering valve **110** outputs the hydraulic fluid through the retraction conduit **111**, thereby causing hydraulic fluid to flow into the rod ends **122**, **132**, **142**, **152** of the respective cylinders. As a result, the respective piston rods **128**, **136**, **146**, **156** retract, and hydraulic fluid from the cap ends **124**, **134**, **144**, **154** is received by the extension conduit **113**.

Moreover, retraction of the piston rods of the respective cylinders drives the wheel assemblies to move to a lower vertical position.

[0051] Furthermore, by providing hydraulic fluid to the second hydraulic fluid input **104**, the hydraulic fluid flows through the second supply conduit **108** to the main lift metering valve **110**. The main lift metering valve **110** outputs the hydraulic fluid through the extension conduit **113**, thereby causing hydraulic fluid to flow into the cap ends **124**, **134**, **144**, **154** of the respective cylinders. As a result, the respective piston rods **128**, **136**, **146**, **156** extend, and hydraulic fluid from the rod ends **122**, **132**, **142**, **152** is received by the retraction conduit **111**. Moreover, retraction of the piston rods of the respective cylinders drives the wheel assemblies to move to a higher vertical position. In certain embodiments, providing hydraulic fluid to the first hydraulic fluid input **102** may cause the main lift metering valve **110** to output hydraulic fluid through the extension conduit **113**, and providing hydraulic fluid to the second hydraulic fluid input **104** may cause the main lift metering valve **110** to output hydraulic fluid through the retraction conduit **111**.

[0052] Additionally, a hydraulic pump **158** is located on the work vehicle (e.g., tractor) that tows the tillage implement. The hydraulic pump **158** is configured to receive hydraulic fluid from a tank **160** and to provide the fluid to the main lift metering valve **110** via a supply line **164**. Further, the tank **160** is configured to receive hydraulic fluid from the main lift metering valve **110** via a return line **162** that fluidly connects the tank **160** to the main lift metering valve **110**.

[0053] In certain embodiments, the controller controls the main lift metering valve **110** to fluidly connect the supply line **164** to the retraction conduit **111**, and to fluidly connect the return line **162** to the extension conduit **113**. In this embodiment, the controller controls the pump **158** to introduce hydraulic fluid to the retraction conduit **111**, via the main lift metering valve **110**, via the supply line **164**. As a result, the controller may control the vertical positions of the wheel assemblies substantially in unison, driving the piston rods **128**, **136**, **146**, **156** to retract. Further, hydraulic fluid returns to the main lift metering valve via the extension conduit **113**, and then returns to the tank **160** via the return line **162**. In other embodiments, the controller may control the main lift metering valve **110** to fluidly connect the supply line **164** to the extension conduit **113**, and to fluidly connect the return line **162** to the retraction conduit **111**. In this embodiment, the controller controls the pump **158** to introduce hydraulic fluid to the extension conduit **113**, via the main lift metering valve **110**, via the supply line **164**. As a result, the controller may control the vertical positions of the wheel assemblies substantially in unison, driving the piston rods **128**, **136**, **146**, **156** to extend. Further, hydraulic fluid returns to the main lift metering valve **110** via the retraction conduit **111**, and then returns to the tank **160** via the return line **162**.

[0054] In the illustrated embodiment, the hydraulic circuit **100** includes a right wing control valve circuit **180** configured to control the right wing lift cylinder **120** independently of the other cylinders, and the hydraulic circuit **100** includes a left wing control valve circuit **220** configured to control the left wing lift cylinder **150** independently of the other cylinders. The right wing control valve circuit **180** receives hydraulic fluid from the hydraulic pump **158** via a right wing circuit pump conduit **166**. The right wing circuit pump conduit **166** fluidly couples the hydraulic pump **158** and the right wing control valve circuit **180**. Additionally, a right wing circuit tank conduit **168** fluidly couples the tank **160** and the right wing control valve circuit **180**. In addition, the left wing control valve circuit **220** receives hydraulic fluid from the hydraulic pump **158** via a left wing circuit pump conduit **170**. The left wing circuit pump conduit **170** fluidly couples the hydraulic pump **158** and the left wing control valve circuit **220**. Additionally, a left wing circuit tank conduit **172** fluidly couples the tank **160** and the left wing control valve circuit **220**.

[0055] In the illustrated embodiment, the right wing control valve circuit **180** includes a first control valve **181** configured to facilitate the flow of hydraulic fluid into appropriate fluid conduits for adjusting the vertical position of the right wing wheel assembly. In the illustrated embodiment, the control valve **181** is a four-way, three-position solenoid activated valve. In the illustrated embodiment, the control valve **181** includes a first position **182**, a second position **188**, and a third position **190**. With the control valve **181** in the first position **182**, hydraulic fluid from the hydraulic pump **158** is directed along a first valve output conduit **194**, through a first check valve **200**, and through a second valve output conduit **204**. Additionally, with the control valve **181** in the first position **182**, a third valve output conduit **196** is fluidly connected to the right wing circuit tank conduit **168**, enabling flow from the third valve output conduit **196** to return to the tank. The first check valve **200** is configured to enable flow in one direction and normally block flow in the other direction. In the illustrated embodiment, the first check valve **200** is configured to enable flow from the first valve output conduit **194** to the second valve output conduit **204**, and normally block flow of fluid from the second valve output conduit **204** to the first valve output conduit **194**. As discussed in more detail below, the first check valve **200** is configured to open and enable flow from the second valve output conduit **204** to the first valve output conduit **194** when a first pilot conduit **206** is pressurized, thereby opening the first check valve **200**. The right wing control valve circuit **180** includes a second check valve **198** configured to enable flow in one direction and normally block flow in the other direction. In the illustrated embodiment, the second check valve **198** is configured to enable flow from the third valve output conduit **196** to a fourth valve output conduit **202**, and normally block flow of fluid from the fourth valve output conduit **202** to the third valve output conduit **196**. As discussed in more detail below, the second check valve **198** is configured to open and enable hydraulic fluid to flow from the fourth valve output conduit **202** to the third valve output conduit **196** when a second pilot conduit **208** is pressurized, thereby opening the second check valve **198**.

[0056] The second valve output conduit **204** is fluidly coupled to the rod end **122** of the right wing lift cylinder **120**. By introducing hydraulic fluid to the rod end **122** of the right wing lift cylinder **120**, as discussed previously, the piston rod **128** is driven to retract, thereby decreasing the vertical position of the corresponding right wing wheel assembly. As the cylinder retracts, hydraulic fluid is driven from the cap end **124** into the fourth valve output conduit **202**. The hydraulic fluid is driven into the fourth valve output conduit **202**, rather than the rod end **132** of right main lift cylinder **130**, because the right main lift cylinder **130** is blocked from receiving hydraulic fluid as the main lift metering valve **110**

is not driving this fluid motion. Because the control valve **181** is in the first position **182**, the second pilot conduit **208** is pressurized, thereby enabling the hydraulic fluid to pass through the second check valve **198**, through the third valve output conduit **196**, and return back to the tank **160** through the right wing circuit tank conduit **168**.

[0057] When the control valve **181** is in the second position **188**, the control valve **181** blocks hydraulic fluid from flowing from the hydraulic pump **158** into any of the above mentioned valve output conduits **194**, **204**, **196**, **202**. In addition, when the control valve **181** is in the second position **188**, a fluid connection between the second valve output conduit **204** and the right wing circuit tank conduit **168** and a fluid connection between the fourth valve output conduit **202** and the right wing circuit tank conduit **168** are established. Accordingly, the first pilot conduit **206** and the second pilot conduit **208** are not pressurized. As a result, fluid flow from the cap end **124** and the rod end **122** of the right wing lift cylinder **120** is blocked, thereby blocking movement of the piston rod **128** (e.g., while hydraulic fluid is not provided by the main lift metering valve **110**). In the illustrated embodiment, the control valve **181** includes a first biasing spring **184** and a second biasing spring **192**. The first biasing spring **184** and the second biasing spring **192** bias the control valve **181** to normally be in the second position **188**. The control valve **181** includes a solenoid **186** that is configured to provide a force that overcomes the force of the first biasing spring **184** and the second biasing spring **192**, thereby driving the control valve to the first position **182** or to the third position **190**.

[0058] The solenoid **186** is communicatively coupled to the controller, thereby enabling the controller to control the control valve **181**. Accordingly, the controller may control the right wing lift cylinder **120**. For example, the controller may control the solenoid **186** to drive the control valve **181** to the third position **190**. With the control valve **181** in the third position **190**, a fluid connection is established between the third valve output conduit **196** and the right wing circuit pump conduit **166**, and a fluid connection is established between the first valve output conduit **194** and the right wing circuit tank conduit **168**. Accordingly, hydraulic fluid from the hydraulic pump **158** flows through the third valve output conduit **196**, through the second check valve **198**, and through the fourth valve output conduit **202**. The fourth valve output conduit **202** is fluidly coupled to the cap end **124** of the right wing lift cylinder **120**. By introducing hydraulic fluid to the cap end **124** of the right wing lift cylinder **120**, as discussed previously, the piston rod **128** is driven to extend, thereby increasing the vertical position of the right wing wheel assembly. As the cylinder extends, hydraulic fluid is driven from the rod end **122** into the second valve output conduit **204**. The hydraulic fluid is driven into the second valve output conduit **204**, rather than the right branch retraction conduit **112**, because the right branch retraction conduit **112** is blocked from receiving hydraulic fluid as the main lift metering valve **110** is not driving this fluid motion. Because the control valve **181** is in the third position **190**, the first pilot conduit **206** is pressurized, thereby opening the first check valve **200**. As a result, hydraulic fluid flows through the first check valve **200**, through the first valve output conduit **194**, and returns to the tank **160** through the right wing circuit tank conduit **168**.

[0059] In the illustrated embodiment, the right wing control valve circuit **180** includes a shuttle valve **210** configured to transfer a pressure from the rod end **122** or the cap end **124** of the right wing lift cylinder **120** to a spring end of a compensator **212** disposed along the right wing circuit pump conduit **166**. The shuttle valve **210** and the compensator **212** may be configured to create a substantially constant pressure differential across the control valve **181**, thereby facilitating a fluid flow more linearly related to the opening of the control valve **181** in the various positions. As a result, the consistency of the flow through the control valve **181** relative to the position of the control valve **181** may be enhanced, thereby facilitating controller control of the control valve **181**. In certain embodiments, the shuttle valve and the compensator may be omitted. Furthermore, in certain embodiments, the check valves and the pilot conduits may be omitted. In such embodiments, the control valve may be configured to block flow through the control valve while in the second position. In certain embodiments, the right wing control valve circuit **180** enables the operator to adjust the vertical position of the right wing wheel assembly while the tillage implement is being towed and is in operation.

[0060] In the illustrated embodiment, the left wing control valve circuit **220** includes a control valve **221** configured to facilitate the flow of hydraulic fluid into appropriate fluid conduits for adjusting the vertical position of the left wing wheel assembly. In the illustrated embodiment, the control valve **221** is a four-way, three-position solenoid activated valve. In the illustrated embodiment, the control valve **221** includes a first position **222**, a second position **228**, and a third position **230**. With the control valve **221** in the first position **222**, hydraulic fluid from the hydraulic pump **158** is directed along a fifth valve output conduit **234**, through a third check valve **240**, and through a sixth valve output conduit **244**. Additionally, with the control valve **221** in the first position **222**, a seventh valve output conduit **236** is fluidly connected to the left wing circuit tank conduit **172**, enabling flow from the seventh valve output conduit **236** to return to the tank. The third check valve **240** is configured to enable flow in one direction and normally block flow in the other direction. In the illustrated embodiment, the third check valve **240** is configured to enable flow from the fifth valve output conduit **234** to the sixth valve output conduit **244**, and normally block flow of fluid from the sixth valve output conduit **244** to the fifth valve output conduit **234**. As discussed in more detail below, the third check valve **240** is configured to open and enable flow from the sixth valve output conduit **244** to the fifth valve output conduit **234** when a third pilot conduit **248** is pressurized, thereby opening the third check valve **240**. The left wing control valve circuit **220** includes a fourth check valve **238** configured to enable flow in one direction and normally block flow in the other direction. In the illustrated embodiment, the fourth check valve **238** is configured to enable flow from the seventh valve output conduit **236** to an eighth valve output conduit **242**, and normally block flow of fluid from the eighth valve output conduit **242** to the seventh valve output conduit **236**. As discussed in more detail below, the fourth check valve **238** is configured to open and enable fluid to flow from the eighth valve output conduit **242** to the seventh valve output conduit **236** when a fourth pilot conduit **246** is pressurized, thereby opening the fourth check valve **238**.

[0061] The sixth valve output conduit **244** is fluidly coupled to the rod end **152** of the left wing lift cylinder **150**. By introducing hydraulic fluid to the rod end **152** of the left wing lift cylinder **150**, as discussed previously, the piston rod **156** is driven to retract, thereby decreasing the vertical position of the corresponding left wing wheel assembly. As the cylinder retracts, hydraulic fluid is driven from the cap end **154** and into the eighth valve output conduit **242**. The hydraulic fluid is driven into the eighth valve output conduit **242**, rather than the rod end **142** of left main lift cylinder **140**, because the left main lift cylinder **140** is blocked from receiving hydraulic fluid as the main lift metering valve **110** is not driving this fluid motion. Because the control valve **221** is in the first position **222**, the fourth pilot conduit **246** is pressurized, thereby enabling the hydraulic fluid to pass through the fourth check valve **238**, through the seventh valve output conduit **236**, and return back to the tank **160** through the left wing circuit tank conduit **172**.

[0062] When the control valve **221** is in the second position **228**, the control valve **221** blocks hydraulic fluid from flowing from the hydraulic pump **158** into any of the above mentioned valve output conduits **234**, **244**, **236**, **242**. In addition, when the control valve **221** is in the second position **228**, a fluid connection between the sixth valve output conduit **244** and the left wing circuit tank conduit **172** and a fluid connection between the eighth valve output conduit **242** and the left wing circuit tank conduit **172** is established. Accordingly, the third pilot conduit **248** and the fourth pilot conduit **246** are not pressurized. As a result, fluid flow from the cap end **154** and the rod end **152** of the left wing lift cylinder **150** is blocked, thereby blocking movement of the piston rod **156** (e.g., while hydraulic fluid is not provided by the main lift metering valve **110**). In the illustrated embodiment, the control valve **221** includes a third biasing spring **224** and a fourth biasing spring **232**. The third biasing spring **224** and the fourth biasing spring **232** bias the control valve **221** to normally be in the second position **228**. The control valve **221** includes a solenoid **226** that is configured to provide a force that overcomes the force of the third biasing spring **224** and the fourth biasing spring **232**, thereby driving the control valve **221** to the first position **222**, or to the third position **230**.

[0063] The solenoid **226** is communicatively coupled to the controller, thereby enabling the controller to control the control valve **221**. Accordingly, the controller may control the left wing lift cylinder **150**. For example, the controller may control the solenoid **226** to drive the control valve **221** to the third position **230**. With the control valve **221** in the third position **230**, a fluid connection is established between the seventh valve output conduit **236** and the left wing circuit pump conduit **170**, and a fluid connection is established between the fifth valve output conduit **234** and the left wing circuit tank conduit **172**. Accordingly, hydraulic fluid from the hydraulic pump **158** flows through the seventh valve output conduit **236**, through the fourth check valve **238**, and through the eighth valve output conduit **242**. The eighth valve output conduit **242** is fluidly coupled to the cap end **154** of the left wing lift cylinder **150**. By introducing hydraulic fluid to the cap end **154** of the left wing lift cylinder **150**, as discussed previously, the piston rod **156** is driven to extend, thereby increasing the vertical position of the left wing wheel assembly. As the cylinder extends, hydraulic fluid is driven from the rod end **152** and into the sixth valve output conduit **244**. The hydraulic fluid is driven into the sixth valve output conduit **244**, rather than the left branch retraction conduit **114**, because the left branch retraction conduit **114** is blocked from receiving hydraulic fluid as the main lift metering valve **110** is not driving this fluid motion. Because the control valve **221** is in the third position **230**, the third pilot conduit **248** is pressurized, thereby opening the third check valve **240**. As a result, hydraulic fluid flows through the third check valve **240**, through the fifth valve output conduit **234**, and returns to the tank **160** through the left wing circuit tank conduit **172**.

[0064] In the illustrated embodiment, the left wing control valve circuit **220** includes a shuttle valve **250** configured to transfer a pressure from the rod end **152** or the cap end **154** of the left wing lift cylinder **150** to a spring end of a compensator **252** disposed along the left wing circuit pump conduit **170**. The shuttle valve **250** and the compensator **252** may be configured to create a substantially constant pressure differential across the control valve **221**, thereby facilitating a fluid flow more linearly related to the opening of the control valve **221** in the various positions. As a result, the consistency of the flow through the control valve **221** relative to the position of the control valve **221** may be enhanced, thereby facilitating controller control of the control valve **221**. In certain embodiments, the shuttle valve and the compensator may be omitted. Furthermore, in certain embodiments, the check valves and the pilot conduits may be omitted. In such embodiments, the control valve may be configured to block flow through the control valve while in the second position. In certain embodiments, the left wing control valve circuit **220** enables the operator to adjust the vertical position of the left wing wheel assembly while the tillage implement is being towed and is in operation. In certain embodiments, the right wing lift cylinder **120** and the left wing lift cylinder **150** may be manually actuated, rather than being controlled by the controller. In other embodiments, extension of the piston rods **128**, **136**, **146**, **156** may drive the wheel assemblies upwards towards the frame, while retraction of the piston rods **128**, **136**, **146**, **156** may drive the wheel assemblies downwards away from the frame.

[0065] While only certain features have been illustrated and described herein, many modifications and changes will occur to those skilled in the art. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the true spirit of the disclosure.

[0066] The techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field and, as such, are not abstract, intangible or purely theoretical. Further, if any claims appended to the end of this specification contain one or more elements designated as "means for (perform)ing (a function) . . . " or "step for (perform)ing (a function) . . . ", it is intended that such elements are to be interpreted under 35 U.S.C. 112(f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. 112(f).

1. A height adjustment system for an agricultural implement, comprising:

a main lift cylinder configured to control a vertical position of a center section wheel assembly, wherein the center section wheel assembly is movably coupled to a center section of a frame of the agricultural implement;

a wing lift cylinder configured to control a vertical position of a wing section wheel assembly, wherein the wing section wheel assembly is movably coupled to a wing section of the frame of the agricultural implement; and

a hydraulic circuit comprising:

a main lift metering valve configured to control hydraulic fluid flow to and from the main lift cylinder and the wing lift cylinder to control the vertical position of the center section wheel assembly and the vertical position of the wing section wheel assembly in unison; and

a wing control valve circuit comprising a first control valve, wherein the first control valve is configured to control fluid flow to and from the wing lift cylinder to control the vertical position of the wing section wheel assembly independently of the center section wheel assembly.

2. The height adjustment system of claim **1**, comprising:

an additional wing lift cylinder configured to control an additional vertical position of an additional wing section wheel assembly, wherein the additional wing section wheel assembly is movably coupled to an additional wing section of the frame of the agricultural implement; and

wherein the hydraulic circuit comprises an additional wing control valve circuit comprising a second control valve is configured to control fluid flow to and from the additional wing lift cylinder to control the additional vertical position of the additional wing section wheel assembly independently of the center section wheel assembly.

3. The height adjustment system of claim **1**, wherein the first control valve is a four-way, three-position, solenoid activated valve.

4. The height adjustment system of claim **2**, wherein the second control valve is a four-way, three-position, solenoid activated valve.

5. The height adjustment system of claim **1**, comprising:

a control system comprising a controller having a processor and a memory, wherein the controller is configured to:

receive from a user interface a first input corresponding to a first adjustment in the vertical position of the wing section wheel assembly; and

control the first control valve based on the first input received from the user interface.

6. The height adjustment system of claim **1**, wherein the wing control valve circuit comprises:

a compensator with a spring end, wherein the compensator is disposed along an input conduit to the wing control valve circuit; and

a shuttle valve configured to transfer a pressure from a rod end or a cap end of the wing lift cylinder to the spring end of the compensator, wherein the shuttle valve and the compensator are configured to create a substantially constant pressure differential across the first control valve.

7. The height adjustment system of claim **1**, wherein the wing control valve circuit comprises:

a check valve configured to enable fluid flow from a first output valve conduit to a second output valve conduit, and normally block fluid flow from the second output valve conduit to the first output valve conduit; and

a pilot conduit fluidly coupled to the check valve, wherein the check valve is configured to open and enable fluid to flow from the second output valve conduit to the first output valve conduit in response to the pilot conduit being pressurized.

8. The height adjustment system of claim **1**, wherein the hydraulic circuit comprises:

a first hydraulic fluid input configured to provide hydraulic fluid through a first supply conduit to the main lift metering valve; and

a second hydraulic fluid input configured to provide hydraulic fluid through a second conduit to the main lift metering valve.

9. An agricultural implement, comprising:

a frame comprising a center section, a left wing section, and a right wing section;

a center section wheel assembly movably coupled to the center section;

a right wing wheel assembly movably coupled to the right wing section;

a left wing wheel assembly movably coupled to the left wing section;

a height adjustment system comprising:

a main lift cylinder configured to control a first vertical position of the center section wheel assembly;

a right wing lift cylinder configured to control a second vertical position of the right wing wheel assembly;

a left wing lift cylinder configured to control a third vertical position of the left wing wheel assembly; and

a hydraulic circuit comprising:

a main lift metering valve configured to control hydraulic fluid flow to and from the main lift cylinder, the right wing lift cylinder, and the left wing lift cylinder to control the first vertical position of the center section wheel assembly, the second vertical position of the right wing wheel assembly, and the third vertical position of the left wing wheel assembly in unison;

a right wing control valve circuit comprising a first control valve, wherein the first control valve is configured to control fluid flow to and from the right wing lift cylinder to control the second vertical position of the right wing wheel assembly independently of the center section wheel assembly; and

a left wing control valve circuit comprising a second control valve, wherein the second control valve is configured to control fluid flow to and from the left wing lift cylinder to control the third vertical position of the left wing wheel assembly independently of the center section wheel assembly.

10. The agricultural implement of claim **9**, comprising:

a control system comprising a controller having a processor and a memory, wherein the controller is configured to:

receive from a user interface a first input corresponding to a first adjustment in the second vertical position of the right wing wheel assembly; and

control the first control valve based on the first input received from the user interface.

11. The agricultural implement of claim **10**, wherein the controller is configured to:

receive from the user interface a second input corresponding to a second adjustment in the third vertical position of the left wing wheel assembly; and control the second control valve based on the second input received from the user interface.

12. The agricultural implement of claim **11**, comprising:

a first position sensor configured to monitor the first vertical position of the center section wheel assembly;

a second position sensor configured to monitor the second vertical position of the right wing wheel assembly; and

a third position sensor configured to monitor the third vertical position of the left wing wheel assembly.

13. The agricultural implement of claim **12**, wherein the controller communicatively coupled to the first position sensor, the second position sensor, and the third position sensor.

14. The agricultural implement of claim **13**, wherein the controller is configured to control the hydraulic circuit based on feedback from the first position sensor, the second position sensor, and the third position sensor.

15. A hydraulic circuit, comprising:

a main lift metering valve configured to receive pressurized hydraulic fluid from a hydraulic pump and to output the pressurized hydraulic fluid to raise and lower a height adjustment system for an agricultural implement, wherein the height adjustment system comprises a plurality of actuating cylinders; and

a wing control valve circuit comprising a first control valve, wherein the first control valve is configured to control fluid flow to and from a wing lift cylinder of the plurality of actuating cylinders, wherein the wing lift cylinder is configured to control a vertical position of a wing section wheel assembly independently of a center section wheel assembly.

16. The hydraulic circuit of claim **15**, comprising an additional wing control valve circuit comprising a second control valve, wherein the second control valve is configured to control fluid flow to and from an additional wing lift cylinder of the plurality of actuating cylinders, wherein the additional wing lift cylinder is configured to control an additional vertical position of an additional wing section wheel assembly independently of the center section wheel assembly.

17. The hydraulic circuit of claim **15**, wherein the wing control valve circuit comprises a first four-way, three-position, solenoid activated valve.

18. The hydraulic circuit of claim **15**, wherein the wing control valve circuit comprises:

a compensator with a spring end, wherein the compensator is disposed along an input conduit to the first control valve; and

a shuttle valve configured to transfer a pressure from a rod end or a cap end of the wing lift cylinder to the spring end of the compensator, wherein the shuttle valve and the compensator are configured to create a substantially constant pressure differential across the first control valve.

19. The hydraulic circuit of claim **15**, wherein the wing control valve circuit comprises:

a check valve configured to enable fluid flow from a first output valve conduit to a second output valve conduit, and normally block fluid flow from the second output valve conduit to the first output valve conduit; and

a pilot conduit fluidly coupled to the check valve, wherein the check valve is configured to open and enable fluid to flow from the second output valve conduit to the first output valve conduit in response to the pilot conduit being pressurized.

20. The hydraulic circuit of claim **15**, comprising:

a first hydraulic fluid input configured to provide hydraulic fluid through a first supply conduit to the main lift metering valve; and

a second hydraulic fluid input configured to provide hydraulic fluid through a second conduit to the main lift metering valve.

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